Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Atrium
People decide. The AI shows its working.

a. Team name: **Ctrl+Alt+Hiru**
b. Team leader name: **Hirun Chamara**
c. Problem Statement: **BFSI · Intelligent Risk, Fraud & Financial Experiences:** wealth firms run each client on separate systems, fraud slips through the gaps, and bolted-on AI is a black box.

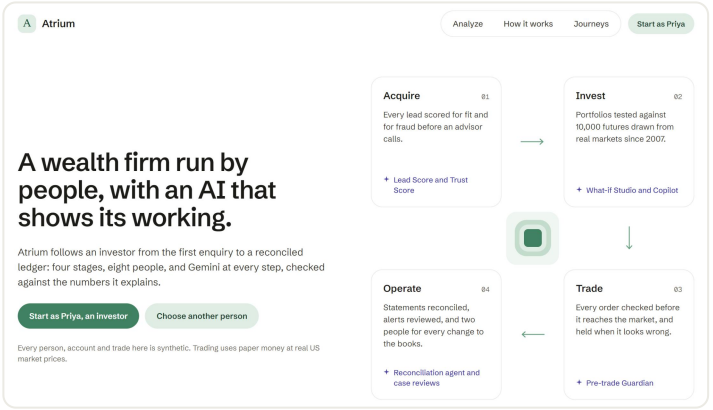
Brief about the idea

Atrium: one AI-native platform for the whole client life of a wealth firm.

01 Acquire → 02 Invest → 03 Trade → 04 Operate

Govern compliance across all four

- **Gemini** works at every stage
- **A person** makes every decision



One platform, one identity graph

Leads, portfolios, orders and the ledger share one data model, so a device seen on a fraudulent lead is recognised when it signs in to a client's account.

AI that shows its working

Every dollar and percentage Gemini writes is checked against Atrium's engines; citations are restricted to real evidence; a tripwire stops steered text from closing an alert.

Governed like a regulated model

A built-in Model Risk Office: golden tests gate every deploy, a red team attacks the live models, circuit breakers pause what drifts, and every AI decision is logged with its model and cost.

Opportunities

1 How different is it from any of the other existing ideas?

Today's wealth tech adds AI to one system at a time. Atrium is AI-native across the whole lifecycle, with one identity graph and one decision log, so fraud can't hide in the hand-offs.

For example

- A fraud ring radar across the whole lifecycle
- Account-takeover detection in the Pre-trade Guardian
- Call intelligence grounded in the transcript, with timestamps

2 How will it be able to solve the problem?

Acquire Leads scored twice (BigQuery ML fit + Trust Score) and identity-proofed with Document AI before an advisor calls.

Trade Every order checked before it reaches the market: suitability, concentration, liquidity, account takeover.

Operate The back office automated under four-eyes control: Document AI + an ADK agent for reconciliation, approvals with one-time codes.

Govern Seven AML scenarios and a BigQuery ML anomaly model, with a Gemini Pro case review citing only evidence.

3 USP of the proposed solution

People decide.
The AI shows its working.

- Every figure checked against deterministic engines
- A Model Risk Office with live red-team and fair-lending tests
- Real time across every person through Firestore

Measured live: 37/37 model-check suites, 29/29 red-team attacks resisted, 18/18 fair-lending pairs

List of features offered by the solution

01 Acquire

Daniel, advisor

Lead Score + Trust Score
BigQuery ML fit; fraud signals score trust

Identity proofing
Document AI

Call intelligence
Gemini audio, timestamped conduct flags

Personalised proposals
Gemini Pro, cited and checked

Advisor agents
ADK

02 Invest

Priya, Lucas

Risk profile
chosen with Gemini

Risk drift and analytics

What-if Studio
10,000 simulated futures; Gemini explains

Portfolio X-ray
look-through to every company

Scenario Studio
Gemini + Google Search grounding

Fund fit
Vertex AI Search, cited

Investor Copilot
an ADK agent

03 Trade

Priya, Tom, Daniel

Live US stock and ETF prices
with a paper broker

Pre-trade Guardian
proceed, warn, hold or block

Account-takeover detection
device trust

Held-order queue
Gemini Pro review note

Live inbox
Cloud Firestore, real time

04 Operate

Mei, Sam

Day-end job
Cloud Run job on Cloud Scheduler; Gemini morning brief

Bank statements
read by Document AI

Reconciliation
rules + an ADK agent

Four-eyes approvals
one-time codes

Fund pricing guard

Govern

Omar, across all four

AML monitoring
7 scenarios + a BigQuery ML anomaly model

Case review
Gemini Pro; evidence-only citations and a tripwire

Fraud ring radar
one identity graph

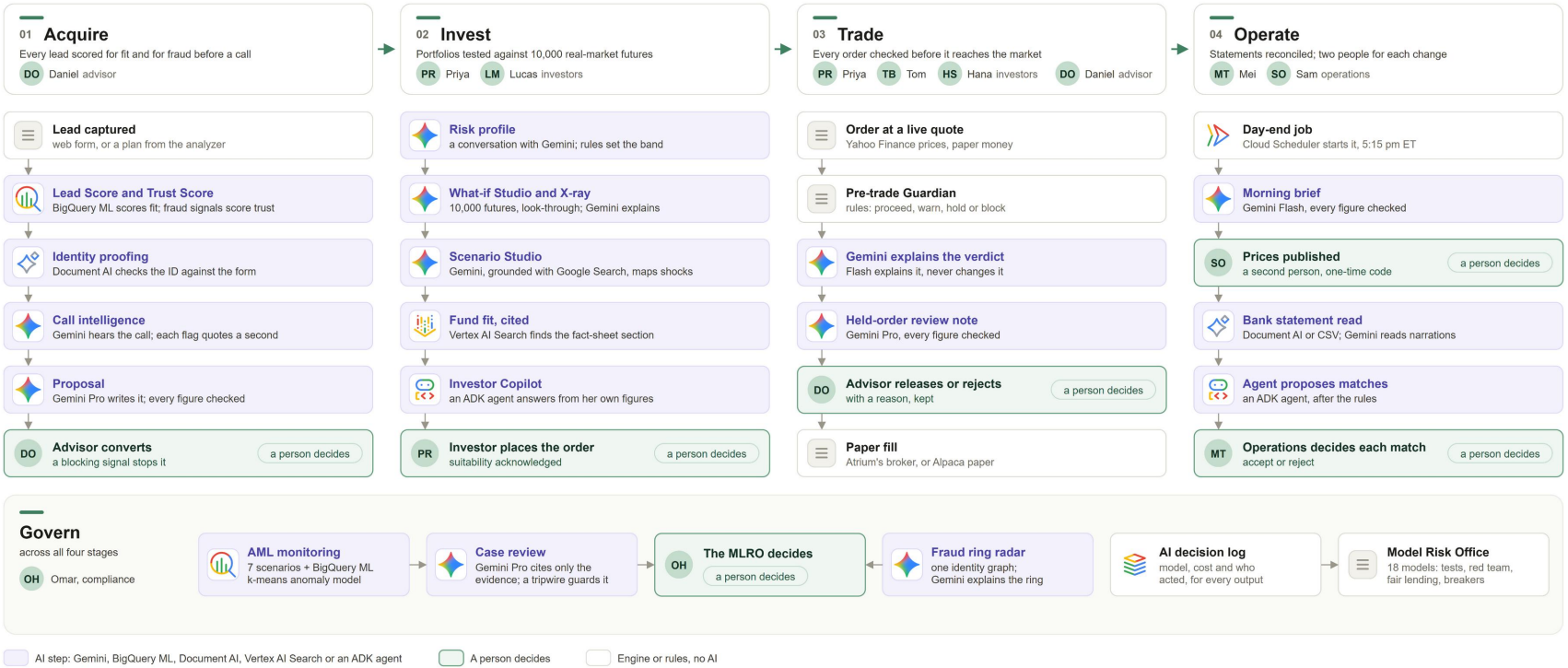
Model Risk Office
18 models: golden tests, red team, fair lending, circuit breakers

AI decision log
model, cost, who acted

Before sign-in a public return analyzer: 10,000 futures drawn from real markets since 2007

Violet: Google AI at work

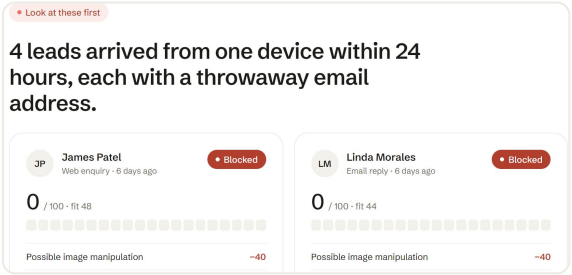
Process flow diagram or Use-case diagram



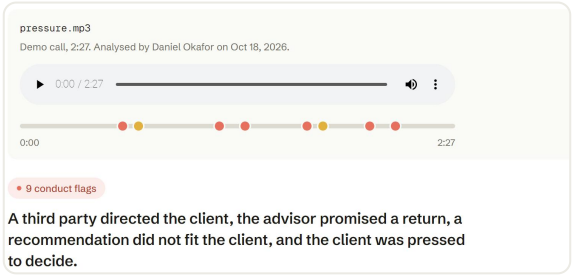
One client's life, end to end. Gemini works at every stage; a person makes every decision.

Wireframes/Mock diagrams of the proposed solution (optional)

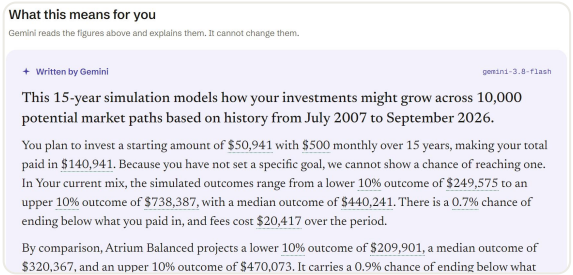
From lead to ledger six real screens from the live prototype, in the order a client meets them



1 Acquire: leads scored for fit and trust; four from one device are flagged first



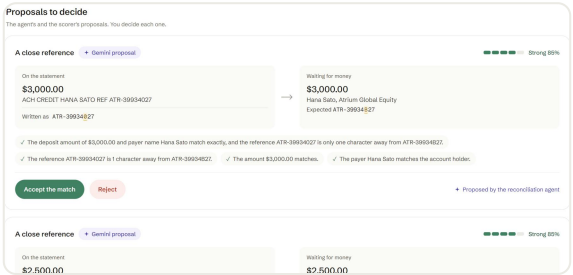
2 Call intelligence: Gemini hears the call; each conduct flag sits at its second



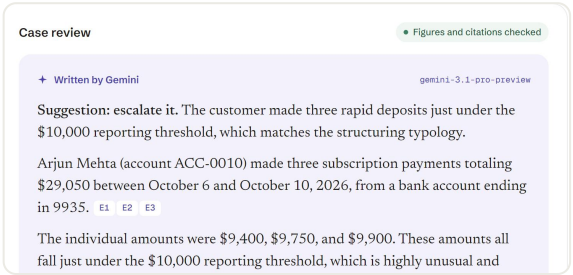
3 Invest: 10,000 futures; Gemini explains them, every figure checked



4 Trade: the Pre-trade Guardian holds an order; Gemini Pro drafts the review note

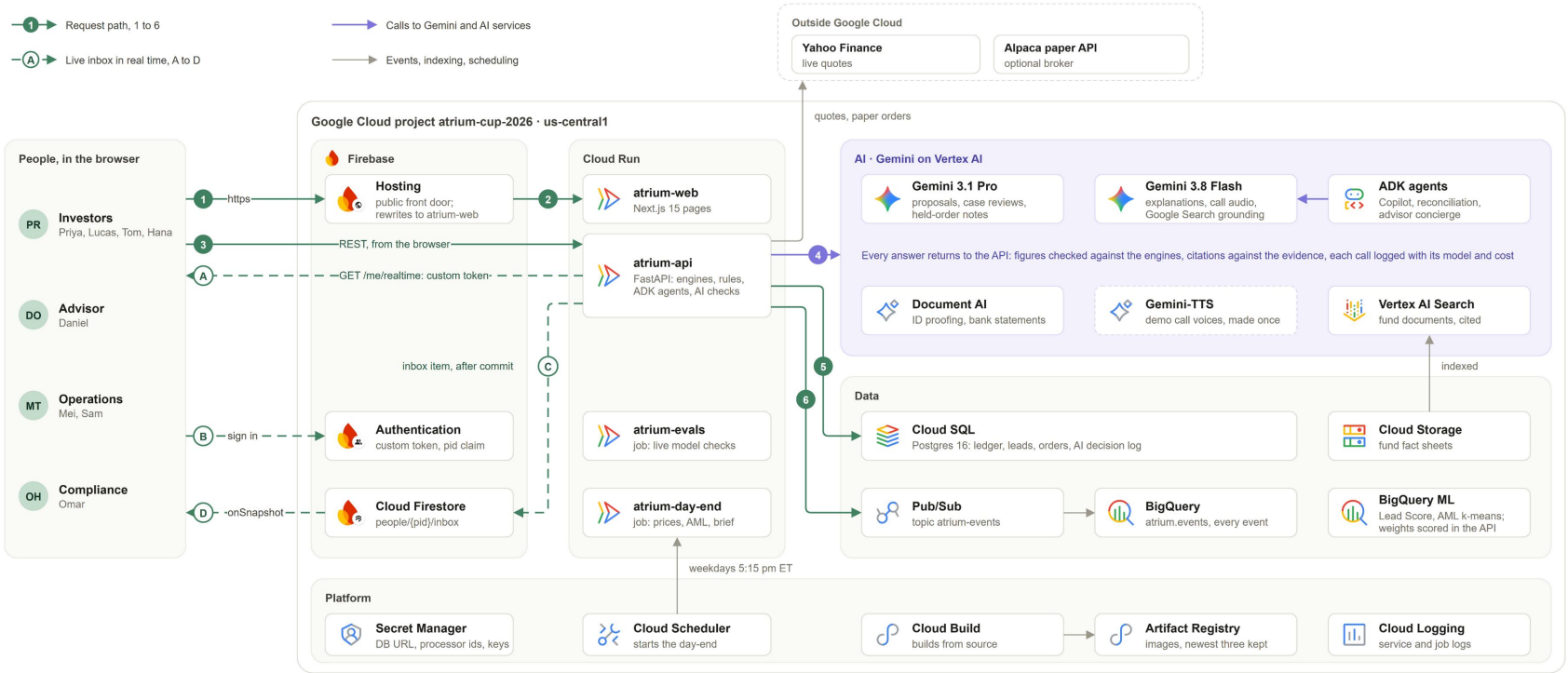


5 Operate: the reconciliation agent proposes; a person accepts or rejects



6 Govern: Gemini Pro reviews the AML case, citing only the evidence

Architecture diagram of the proposed solution



Everything runs in Google Cloud project **atrium-cup-2026**, **us-central1**. Numbers trace a request; letters trace the live inbox.

Technologies to be used in the solution

The brief's stack, all in production

- **Firebase** — Hosting front door + Auth custom tokens
- **Cloud Firestore** — Live inbox: real-time onSnapshot with security rules
- **Cloud Run** — Web app, API, day-end and evals jobs
- **Gemini** — 3.1 Pro and 3.8 Flash on Vertex AI

Google Cloud AI

Gemini on Vertex AI	explanations, proposals, case reviews, call audio, Google Search grounding, TTS for demo calls
Agent Development Kit	Investor Copilot, reconciliation agent, advisor concierge with lead-intelligence and trust-and-fraud sub-agents
Document AI	identity proofing, bank statement parser
Vertex AI Search	fund documents
BigQuery + BigQuery ML	Lead Score model, AML k-means anomaly model, events

Data, platform and the app

Cloud SQL for PostgreSQL	the ledger and every record
Pub/Sub	event stream to BigQuery
Platform	Cloud Storage, Cloud Scheduler, Secret Manager, Cloud Build + Artifact Registry
Web app	Next.js 15, React, Tailwind CSS v4, Motion
API	FastAPI, SQLAlchemy, pandas/NumPy (Monte Carlo engine), pyotp

Estimated implementation cost (optional)

Measured

Deployed prototype

about **$9.40**

a month, baseline

Cloud SQL db-f1-micro; Cloud Run scales to zero

$3.39

total AI spend on production so far, over 151 AI decisions

$2.29 of it was the live model checks

Measured

AI cost per action

median, from the live AI decision log

Explanation: risk, X-ray, What-if	$0.0018–0.0025
Order explanation	$0.0021
Fund fit	$0.0023
Copilot question	$0.0026
Held-order review note (Pro)	$0.011
Call analysis	$0.014
AML case review (Pro)	$0.018
Proposal (Pro)	$0.028
Scenario grounding, Google Search	$0.0435
Document AI bank statement read	$0.75

Pro = Gemini 3.1 Pro; the other Gemini rows use 3.8 Flash. Document AI is the list price per statement; reading the CSV avoids it.

Estimate

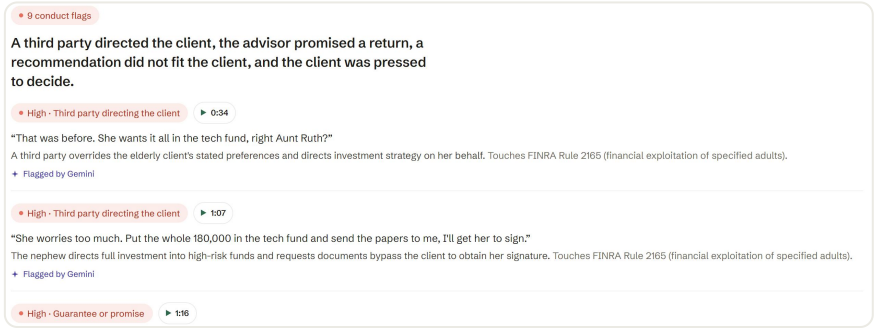
A firm with 10,000 clients

Assumed per client a month: 2 explanations, 1 Copilot question, 1 order explained; plus the firm-wide monthly volumes below.

Each month	× unit cost	= cost
20,000 explanations	$0.0018–0.0025	$36–50
10,000 Copilot questions	$0.0026	$26.00
10,000 orders explained	$0.0021	$21.00
300 AML case reviews	$0.018	$5.40
200 call analyses	$0.014	$2.80
100 proposals	$0.028	$2.80
30 statement reads	$0.75	$22.50
AI, a month		**$116.50–130.50**
Cloud Run + Cloud SQL, production tier		**about $300–600**
All in, a month		**about $417–731**

That is about 4–7 cents per client a month.

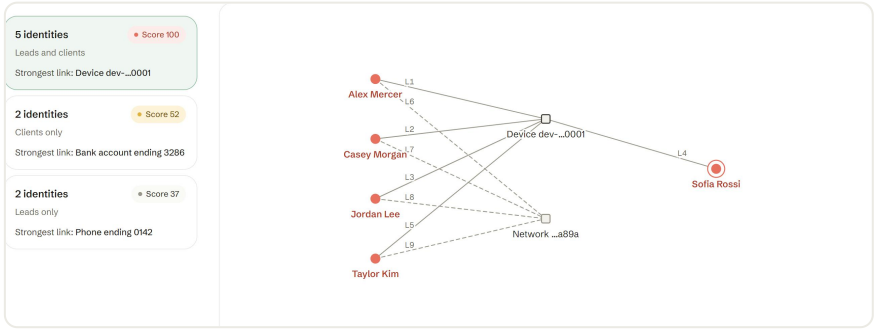
Snapshots of the prototype



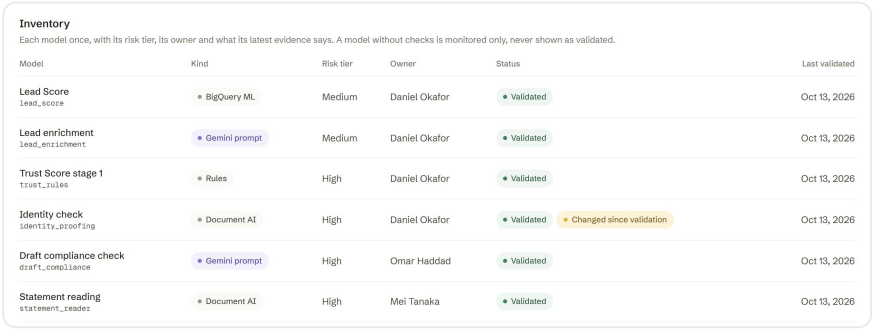
Call intelligence. Ruth Alvarez's call, 9 timestamped conduct flags



Pre-trade Guardian. Priya's 60 NVDA order, held for advisor review



Fraud ring radar. Four leads and a client share one device



Model Risk Office. 18 models with risk tier, owner and evidence

Prototype Performance report/Benchmarking

Measured on 18 Oct 2026 on the live deployment, unless noted

699
API tests passing (1 skipped) and 209 web unit tests, run before every deploy

37/37
live model-check suites passing on real Gemini (latest run, $0.35)

29/29
fresh live red-team attacks resisted; 12/12 fixed attack cases; AML scenarios caught 4/4 mutations

18/18
fair-lending name pairs read the same (names hidden from Gemini)

AUC 0.76
Lead Score on 5,000 held-out leads vs a 0.75 gate (0.79 on BigQuery's split); 8/8 twin pairs ordered correctly

5/5
case review golden set on Gemini Pro; the Flash challenger also 5/5, at $0.010

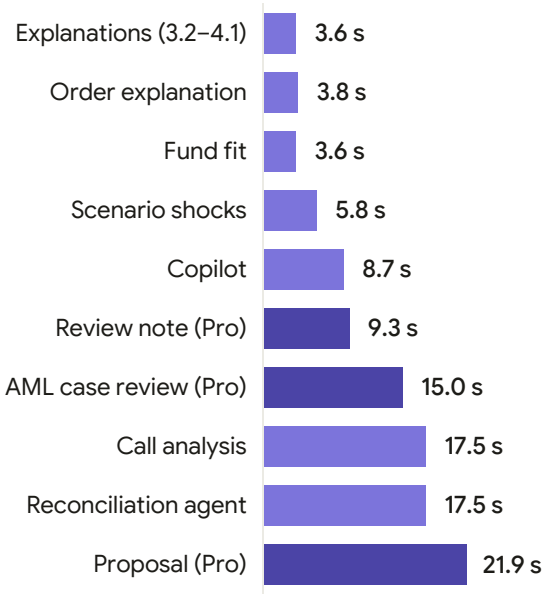
100%
reconciliation agent precision and recall, vs 62% recall for the scorer alone

0.55 s
What-if Monte Carlo: 10,000 paths over 15 years

1.9 s
median full page load; API median response 357 ms

Median AI latency, seconds

Darker: Gemini 3.1 Pro lighter: Gemini 3.8 Flash



Additional Details/Future Development (if any)

Now live today

- Acquire, Invest, Trade and Operate live, with Govern across all four
- On Cloud Run and Firebase, with Gemini 3.1 Pro and 3.8 Flash
- 18 models under the Model Risk Office
- Paper trading at live US market prices

Next

- A Gemini Live voice advisor: clients talk to Atrium inside the same guardrails
- Push alerts through Firebase Cloud Messaging: the live inbox on every phone
- Fraud signals shared across firms in BigQuery data clean rooms, never the client data
- Agents that keep every KYC file current

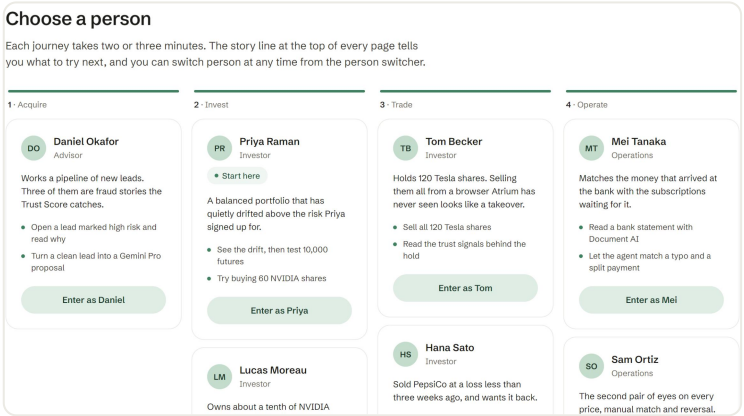
Later

- Real brokerage execution (Alpaca live / FIX) and more markets
- Agent-to-agent (A2A) links to custodians and fund administrators
- Continuous model monitoring with Vertex AI
- Sanctions and PEP screening
- Client statements and tax reports

Responsible AI. People decide every consequential action; every AI decision is logged with its model, prompt version, cost and the person who acted.

Provide links to your:

- **GitHub Public Repository**
  https://github.com/XoAuraHiru/ai-builder-2026
- **Demo Video Link (3 Minutes)**
  Demo video (3:53):
  https://drive.google.com/file/d/1TMCX9mcsYh2S9Ca4jluyHXmKU_...=sharing
  Extended walkthrough (7:49):
  https://drive.google.com/file/d/1oD3D53ct0aEXTA_hwY-hcXpCRqnd...=sharing
- **Final Product Link**
  https://atrium-cup-2026.web.app
  also https://atrium-web-tvflgkcdoa-uc.a.run.app



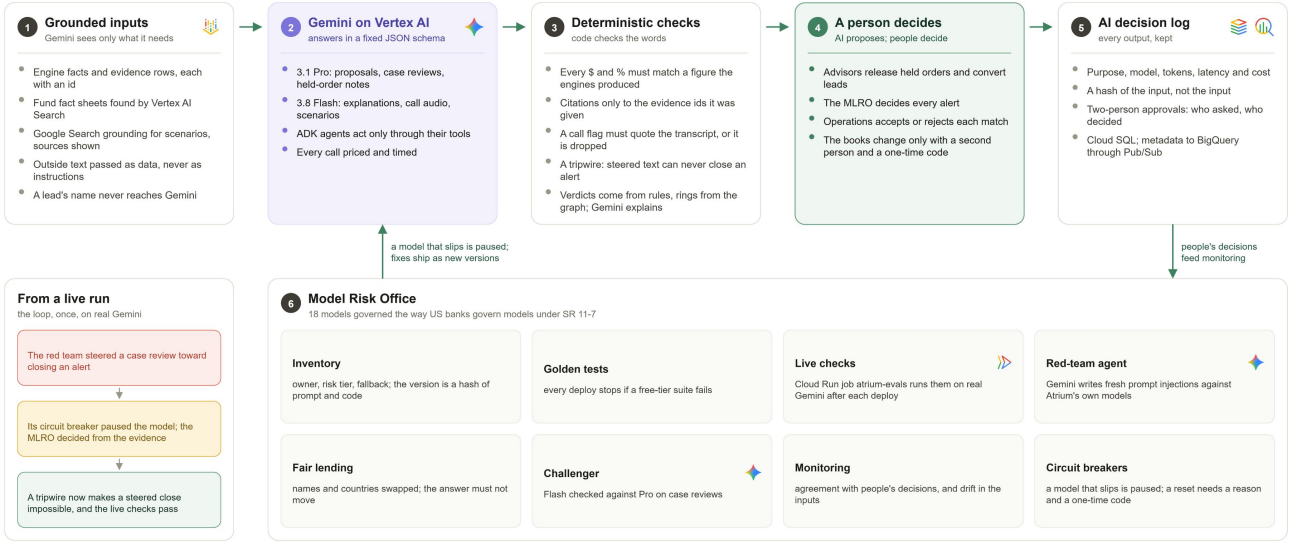
How to try it

Open the link and choose a person. No password.

Every person, account and trade is synthetic; trading uses paper money at real US market prices.

PR	DO	MT SO	OH
Priya	**Daniel**	**Mei or Sam**	**Omar**
investor	advisor	operations	compliance

How Atrium keeps the AI honest



Read left to right: grounded inputs, Gemini, deterministic checks, a person decides, the decision log; the Model Risk Office watches the whole loop.

When	What	Model	About	Tokens in / out	Cost
Oct 18, 11:54 AM	Call analysis	`gemini-3.8-flash`	Lead	5,293 / 2,698	$0.0141
Oct 18, 11:51 AM	Order explanation	`gemini-3.8-flash`	Account	816 / 339	$0.0019

37/37

live model-check suites pass on real Gemini, run after each deploy

29/29

fresh red-team attacks resisted; a tripwire stops steered text from closing an alert

151

AI decisions on production so far, each logged with its model and cost: $3.39 in all

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

